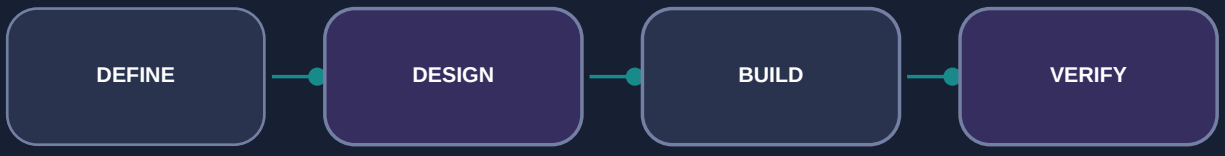
DOCUMENT PLAN • Implementation Plan

Implementation Plan

Milestones, implementation tasks, QA, deployment and traceability



Project name: Slotwise (provisional)

Proposed stack: React + Vite + Tailwind • Node.js + Express • Vercel

Public-source clean-room plan • illustrative graphics, not product screenshots

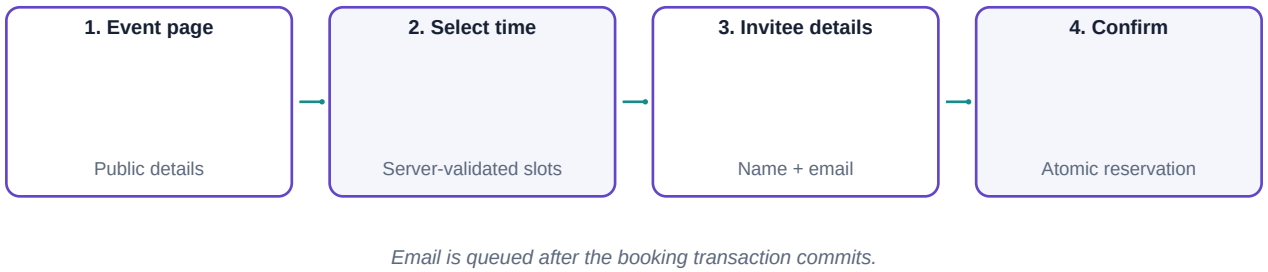
Visual summary

Implementation Plan

PURPOSE

Sequence prototype, implementation, verification and release with measurable gates.

KEY VISUAL



Email is queued after the booking transaction commits.

REVIEW NOTE

Original content is retained after this summary; proposed visuals are illustrative, not real product screenshots.

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Implementation Plan

Milestones, QA, parity, deployment, launch and traceability

Project: Slotwise — provisional independent scheduling product

Reference: Calendly public website and help documentation; no private access.

Platform: Web | **Proposed frontend:** React + Vite + Tailwind CSS | **Backend:** Node.js + Express

Hosting preference: Vercel | **Database:** unresolved conflict between MongoDB preference and required PostgreSQL DDL

Inputs used / assumptions

Planning estimates only. The project is a web-based clean-room scheduling MVP called Slotwise provisionally. It uses stable screen IDs S01–S11, flows F00–F03, and feature priorities must/should/could. Database selection and customer persona require confirmation.

Evidence discipline: Verified public capabilities are distinguished from design proposals and guesses. No review corpus, actual application screenshots, private account data, or production credentials were provided.

1. Delivery strategy

Build a working core loop with a fake data layer first, then swap in the real backend. This validates UX and cross-document assumptions before expensive integration. Use a single modular backend and one database.

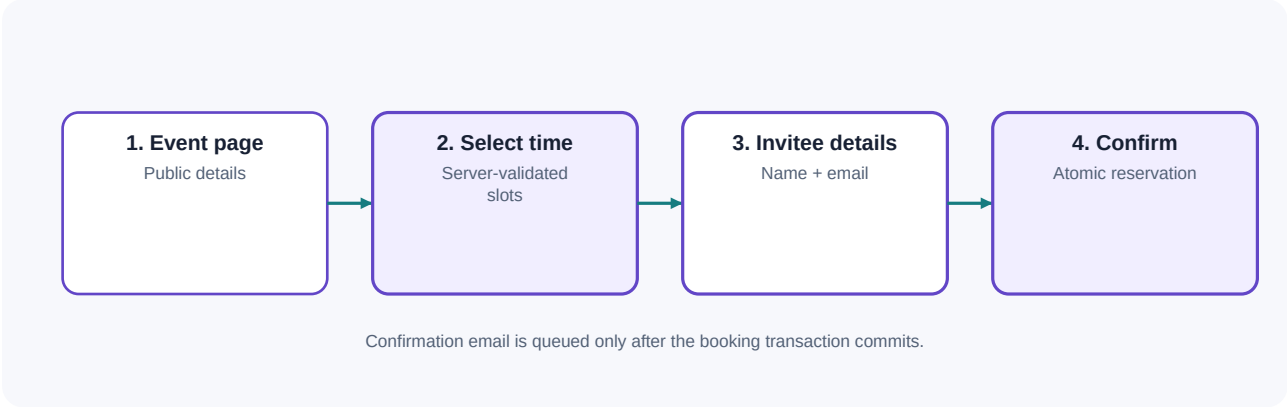

Figure 1. Vertical slice to prove before production integration.

2. Milestones

ID	Milestone	Screens / scope	Tables / routes	Effort	Dependencies / definition of done
M0	Shell	S01–S11 route shells, tokens, primitives, seed data	Mock layer	M	None; all routes render responsively
M1	Vertical slice	S02–S10; F01–F03 with fake layer	Mock signatures for events, slots, booking	L	M0; complete core loop without manual data edits
M2	Must-haves	S01–S09, S11; auth, event CRUD, availability, booking	users, sessions, event_types, availability_rules, bookings; auth/event/public routes	L	M1; persistent core flows pass acceptance tests
M3	Should-haves	S07–S11; cancellation/reschedule, filters	bookings, management_tokens; management routes	M	M2; token and conflict tests pass
M4	Fixes	S03–S10; setup guidance, zone clarity, conflict recovery	Existing tables	M	M2; fixes have measurable acceptance criteria
M5	Backend swap-in	S01–S11; API, DB transaction, outbox	All seven tables; /api/v1 routes	XL	Approve DB choice; concurrency tests pass
M6	Test and bug-fix	All core screens and flows	All APIs/tables	L	M5; all tests pass, no open S1/S2
M7	Parity check	S01–S11 vs selected documented capabilities	Feature register	M	M6; score ≥80% and all musts pass
M8	Rebrand sweep	All screens and assets	Metadata/assets	S	M7; no copied brand assets or copy
M9	Launch assets	Landing page, demo, policy, terms, waitlist	Optional waitlist persistence	M	M8; all claims substantiated
M10	Deploy	Production configuration, DNS, monitoring	Production DB and secrets	M–L	M9; smoke tests and rollback ready

3. Frontend task plan

Task ID	Task	Priority	Milestone	Done when
FE-01	Create React/Vite app and routes	must	M0	All routes load without console errors
FE-02	Implement tokens and accessible primitives	must	M0	Keyboard/focus states documented
FE-03	Build event editor and availability UI	must	M1/M2	Valid and invalid states implemented
FE-04	Build public booking and slot selector	must	M1/M2	Mock flow completes end-to-end
FE-05	Build confirmation and booking management	must/should	M1/M3	Success, conflict and error states work

Task ID	Task	Priority	Milestone	Done when
FE-06	Responsive and accessibility pass	must	M4/M6	390px and 1440px checks pass

4. Backend and data tasks

Task ID	Task	Priority	Milestone	Done when
BE-01	Approve database and migration approach	must	Before M5	PostgreSQL vs MongoDB decision recorded
BE-02	Implement auth and ownership checks	must	M2/M5	Second-user authorization tests pass
BE-03	Implement event and availability APIs	must	M2/M5	Validation and ownership tests pass
BE-04	Implement atomic booking/idempotency	must	M5	Concurrent requests never create overlap
BE-05	Implement management tokens	should	M3/M5	Tokens hashed, expiring and revocable
BE-06	Implement durable email outbox	must	M5	Retries and duplicate handling tested
BE-07	Backups, logging, health and alerts	must	M6/M10	Restore and operational smoke tests pass

5. Timeline and task tracking

Indicative timeline: 6–8 weeks for a focused prototype with consistent effort and an accessible test environment. The estimate depends on the database decision, email setup, time available, and how much rescheduling is included.

Week	Focus	Primary deliverable
1	M0 + product decisions	Shell, token system, approved scope
2	M1	Mock core loop and original wireframes
3	M2 frontend and backend foundation	Event setup and availability
4	M2/M3	Public booking and owner booking list
5	M4/M5	Fixes, database integration, atomic booking
6	M5/M6	Email outbox, integration and E2E tests
7	M6/M7/M8	QA, parity review and rebrand sweep
8	M9/M10	Launch assets, deployment and smoke tests

6. Test strategy and release criteria

Test ID	Type	Scenario	Expected result
F00-H1	Happy	Register, verify and publish	Valid event published
F00-E1	Edge	Required field missing	Inline validation
F00-N1	Negative	Unverified owner publishes	Denied
F01-H1	Happy	Create/publish event	Public event visible
F01-E1	Edge	Invalid availability	Rejected
F01-N1	Negative	Cross-owner edit	403/no data mutation
F02-H1	Happy	Book available slot	One booking persisted
F02-E1	Edge	Concurrent slot conflict	At most one confirmed booking
F02-N1	Negative	Invalid invitee email	Validation error
F02-H2	Happy	Retry idempotent request	No duplicate booking
F03-H1	Happy	Cancel via valid token	Canceled and slot released
F03-E1	Edge	Reschedule to occupied slot	Old booking preserved

Test ID	Type	Scenario	Expected result
F03-N1	Negative	Invalid token	No booking disclosure
F03-H2	Happy	Reschedule valid booking	New slot reserved safely

- Playwright E2E selectors use accessible roles and labels.
- Fail on console errors, uncaught page errors and unexpected 5xx responses.
- Run axe scans on every core screen.
- Test keyboard-only operation and responsive layouts.
- Test DST boundaries, duplicate requests, concurrent bookings and second-user authorization.
- No open S1/S2 bugs before shipping.

7. Parity score

Weights: must = 3, should = 2, could = 1. Full pass = 1.0, partial = 0.5, fail = 0. Score = weighted achieved points divided by weighted total points × 100.

Verdict	Rule
Not shippable	Any must fails, critical security/integrity issue exists, or score <80%
Shippable	Score ≥80%, all musts pass, all release gates pass
Better for target niche	Shippable plus measured improvement in target-user task outcomes

8. Deployment and pre-deploy checklist

- Create separate staging and production databases and run reviewed migrations.
- Configure environment variables through secure hosting settings; keep real values out of .env.example.
- Configure Vercel project, production domain and DNS.
- Set SPF, DKIM and DMARC for email sending.
- Configure OAuth redirect URIs only for approved integrations.
- Keep Stripe in test mode unless explicit live approval is given.
- Configure monitoring, backups and a tested restore path.
- Publish privacy policy and terms; verify account deletion.
- Run production smoke tests and confirm rollback procedure.

9. Tasks only the owner can perform

- Create and own vendor accounts and provide credentials through secure secret management.
- Approve the final product name after dated trademark/domain/store/handle checks.
- Authorize Google/Microsoft OAuth apps and any required provider review.
- Verify domain ownership and email DNS records.
- Explicitly approve any Stripe live-mode activation.
- Approve privacy, retention, terms, and any legally required compliance decisions.

10. Launch plan

Landing page structure: clear headline, three-step explanation, original illustrative demo, feature list, known limitations, privacy/terms, contact and waitlist. Do not use fabricated testimonials, user counts, ratings, or performance claims.

Recruit the first ten real target users through permitted outreach. Observe event setup, link sharing, booking and recovery from conflicts. Record task completion and qualitative feedback; do not present this small sample as statistically representative.

11. Risk register

Risk	Likelihood	Impact	Mitigation / owner
Double booking	Medium	Critical	Database invariant, transaction and concurrency test / engineering
Time-zone and DST errors	Medium	High	UTC + IANA zones and boundary tests / engineering
Weak differentiation	High	High	User interviews and task testing / product
Database mismatch	High until decided	High	Approve PostgreSQL or redesign MongoDB strategy / owner + architect
Email deliverability	Medium	Medium	SPF/DKIM/DMARC, retries, bounce monitoring / operations

12. Final traceability matrix

Feature	Screen	Flow	Table(s)	API route	Milestone	Test
Account registration	S01	F00	users, sessions	POST /auth/register	M2	F00-H1
Authentication guard	S01	F00	users, sessions	POST /auth/login	M2/M5	F00-N1
Event listing	S02	F01	event_types	GET /event-types	M2	F01-H1
Event CRUD	S03	F01	event_types	POST/PATCH /event-types	M2	F01-H1
Availability	S04	F01	availability_rules	PUT /event-types/:id/availability	M2	F01-E1
Publish	S05	F00/F01	event_types	PATCH /event-types/:id	M2	F00-H1
Public event page	S06	F02	event_types	GET /public/:slug	M2	F02-H1
Slot selection	S07	F02	availability_rules, bookings	GET /public/:slug/slots	M2/M5	F02-E1
Invitee details	S08	F02	bookings	POST /public/:slug/bookings	M2	F02-N1
Atomic booking	S08/S09	F02	bookings	POST /public/:slug/bookings	M5	F02-H1
Idempotent booking	S08/S09	F02	bookings	POST /public/:slug/bookings	M5	F02-H2
Confirmation email	S09	F02	outbox_jobs, bookings	POST /public/:slug/bookings	M5	F02-H1
Cancellation	S10	F03	bookings, management_tokens	POST /manage/:token/cancel	M3/M5	F03-H1
Rescheduling	S10/S07	F03	bookings, management_tokens	POST /manage/:token/reschedule	M3/M5	F03-E1
Booking list	S11	F03	bookings	GET /bookings	M2	F03-H1
Independent rebrand	S01–S11	All	N/A	All UI routes	M8	Rebrand sweep

13. Source register and evidence limitations

Source	URL	Use / limitation
Calendly official site	https://calendly.com/	Reference identity; public website
Event type setup	https://help.calendly.com/hc/en-us/articles/360044700074-Create-a-new-event-type?locale=en-us	Public event configuration workflow
Scheduling FAQ	https://calendly.com/help/scheduling-faq	Scheduling concepts and public help
Confirmation page	https://help.calendly.com/hc/en-us/articles/226767207-Display-and-customize-the-event-confirmation-page	Confirmation behavior
Pricing	https://calendly.com/pricing	Reference for future pricing comparison; no claims about user complaints

Evidence status: thin for user complaints, market demand and differentiation. No review corpus, actual screenshots or private account data were supplied. All pain points, pricing, KPI targets and brand candidates are proposals until validated.